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Human-Software Trust Over 45 Days of Simulated Spaceflight

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Abstract

Spaceflight operations are a dynamic, safety-critical environment that relies on automated systems. Crew are expected to trust systems throughout the missions. Within the spaceflight context, it remains unclear how trust in software changes over time, and how measures of trust relate to measures of usability. This study presents initial results that longitudinally assess trust in an extreme environment analogous to spaceflight. Over a simulated spaceflight mission, crew interacted with a software tool designed to enhance crew autonomy by allowing them to manage their own schedule. They filled out a software trust questionnaire three times in total, the last time being post-mission. They also filled out a post-mission multidimensional usability questionnaire. Results suggested that trust increased over time, driven by items on security and familiarity. Post-mission trust was positively correlated with usability dimensions of attractiveness, dependability, and efficiency. Software trust emerges as a dynamic phenomenon that is closely linked with usability.

Keywords

usability, user experience, trust, spaceflight operations, human-computer interaction

Introduction

Trust and usability are critical components of human-computer interaction (HCI). Trust in software tools that support human performance is particularly important in safety-critical settings, such as deep space exploration missions. Trust is thought to modulate the extent to which a user relies on a system to support their tasks (e.g., Lee & See, 2004). Similarly, usability is essential since poor usability is linked to increased error rates (e.g., Fairbanks & Caplan, 2004). Current NASA standards emphasize that software tools should be highly usable before deployment, with usability assessed through quantitative testing using standardized surveys after users interact with the software (NASA, 2010).

However, there is limited research on the relationship between trust and usability in the context of spaceflight missions, and little evidence on how these constructs change over time in an operational setting. In this study, we apply a quantitative, survey-based approach during a 45-day simulated spaceflight mission to answer two main questions. First, does trust in a software tool change over time in an operationally realistic setting, and which items in a standardized trust measure are associated with this change? Second, is post-mission trust correlated with standardized usability scores in a setting with high ecological validity? Our primary objectives are to establish the temporal dynamics of trust and to examine the convergence of trust and usability. Addressing these questions in a spaceflight context clarifies whether human-software trust is a static phenomenon or one that evolves over prolonged use. It also helps determine whether usability and trust, two constructs central to HCI, are interrelated.

Background

Jian et al. (2000) proposed in their landmark work that trust in an automated system could be operationalized as a single score derived from items related to reliability, integrity, confidence, and familiarity. Since its publication, their 12-item quantitative survey has become a widely used tool for assessing trust within human-system interactions, as evidenced by over 2,500 citations as of May 2026. Existing literature has examined how trust changes across single-session experimental blocks (e.g., Onnasch et al., 2023; Yang et al., 2023), as well as over longer periods. For example, Chancey et al. (2025) measured different types of trust before and after 2 days of system interactions. By distinguishing between

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trust subtypes, they were able to show that system exposure increases trust based on what the system does ("performance-based trust") but not necessarily trust based on how the system operates ("process-based trust"). Broadly, this shows that different aspects of trust change over time. A separate recent example, similar to the work presented here, administered relevant items during a multi-day Mars simulation (Vanderdonckt et al., 2025). The authors probed crew members on trust (measuring perceptions of data safety and misuse) and trustworthiness of content (measuring impressions of information quality and reliability). While their results illustrated how system trust can fluctuate across interactions in a spaceflight context, they did not use a multi-item composite measure of overall trust as used in the present study. Therefore, it is crucial to validate the general claim that trust is dynamic (Schaefer et al., 2016) using a widely adopted composite measure such as Jian et al.'s, while also using item-level analysis to identify which specific aspects of trust are most sensitive to change over time. This will help reinforce our understanding of how trust changes over the course of a mission in an operational spaceflight setting.

Marquez et al. (2022) examined trust and usability in a controlled laboratory study, finding that many subscales of a multidimensional usability scale (User Experience Questionnaire; UEQ; Laugwitz et al., 2008) were highly correlated with Jian et al.'s trust composite. This suggests that users are more likely to trust software systems they perceive as usable. However, two limitations constrain Marquez et al.'s findings. First, the study was conducted in a short-term laboratory setting with limited exposure to the software. Thus, it remains unclear whether the coupling between trust and usability persists with realistic, long-term use. Second, several items in the trust composite overlap semantically with UEQ subdimensions. Specifically, trust items assessing dependability and reliability directly mirror the UEQ's Dependability subscale. This raises the possibility that the observed correlations stem from item wording overlap rather than a substantive relationship between constructs. Re-testing these correlations between trust and the UEQ in a new sample, after many days of software use and after trimming semantically overlapping items, would provide a more rigorous test of Marquez et al.'s original findings.

Approach

We analyzed data from 16 participants in a 45-day spaceflight analog at NASA Johnson Space Center that simulates the isolation and confinement of transit to Mars. During the mission, participants repeatedly interacted with a customized scheduling software tool—Playbook (Marquez et al., 2017; Zheng et al., 2023)—designed to support crew task execution, including daily access to procedures and a timeline of activities, as well as crew autonomy through real-time timeline rescheduling. Playbook presents activities on a timeline where users can schedule and reschedule activities based on their preferences and constraints. Constraint checking was automatically performed by the software tool. The interface is used by all crew members, enabling collaborative rescheduling.

Trust in Playbook was measured via the 12-item trust scale proposed by Jian et al. (2000) at three time points: approximately one-third of the way through the mission, approximately two-thirds of the way through the mission, and post-mission. These time points were dictated by the analog as best accommodating the integrated mission schedule. One participant did not complete the trust survey at the post-mission time point; their partial data were retained in all analyses. Items were rated on 7-point ordinal Likert scales, and the item scores were averaged to create the trust composite. During the post-mission period, participants also completed the 26-item UEQ (Schrepp et al., 2014), which divides usability into six subscales: Attractiveness, Perspicuity, Efficiency, Dependability, Stimulation, and Novelty. Extensive consideration of the UEQ in the spaceflight domain is reported in Shelat et al. (2024, 2025).

To assess whether trust changed over time, we fit a linear mixed-effects model with the trust composite score as the dependent variable, measurement order (1–3, treated as a continuous predictor) as a fixed effect, and random intercepts for participants to account for repeated measures. We then applied the same model structure to each of the 12 individual items within the trust scale to identify which aspects of trust drove any observed change. A false discovery rate (FDR) correction was applied to the item-level tests to control for multiple comparisons. To assess the relationship between trust and usability, we qualitatively identified and removed items within the trust composite that were semantically similar to UEQ items. We then repeated the correlation analyses from Marquez et al. (2022) in this new sample with an FDR correction.

Results

Trust composite scores increased over time ($b = 0.19$, $SE = 0.07$, $p = .0102$; Figure 1). At the individual level, 11 of 16 participants (68.75%) showed increases from measurement 1 to measurement 2, 11 of 15 (73.33%) showed increases from measurement 2 to measurement 3, and 12 of 15 (80.00%) showed increases from measurement 1 to measurement 3.

The item-level analysis (Table 1) revealed FDR-corrected significant increases for two specific items in the trust survey: security ($b = 0.48$, $SE = 0.12$, $p = .0003$, p-FDR $= .0036$) and familiarity ($b = 0.34$, $SE = 0.12$, $p = .0060$, p-FDR $= .0360$). No other items showed significant changes after correction.

We next aimed to replicate Marquez et al.'s (2022) correlations between UEQ subscales and trust after removing items that were semantically similar. The UEQ Dependability subscale was too similar to the following three trust items:

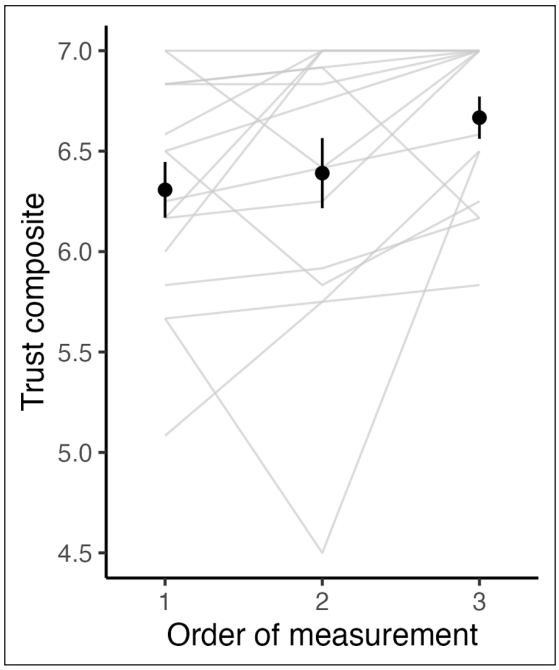


Figure 1. Trust in Playbook increased over time during a 45-day simulated space mission. Points and error bars depict means and standard errors (*SE*), respectively. Light gray lines connect individual participants' data across time.

Table 1. Linear Mixed-Effects Regressions Modeling Item Responses by Measurement Order.

Item	β	*SE*	*p*	*p*-FDR
Security	.48	0.12	.0003	**.0036**
Familiarity	.34	0.12	.0060	**.0360**
Confidence	.38	0.17	.0341	.1284
Trust (single-item)	.25	0.12	.0470	.1284
Reliable	.35	0.17	.0535	.1284
Wary (R)	.09	0.06	.1370	.2740
Harmful (R)	−.03	0.03	.2191	.3588
Deceptive (R)	.12	0.10	.2392	.3588
Integrity	.15	0.16	.3337	.4009
Underhanded (R)	.09	0.09	.3341	.4009
Suspicious (R)	.06	0.08	.4309	.4701
Dependable	−.01	0.14	.9533	.9533

[a]Bolded values indicate significant p-values after correction.

"Playbook is dependable," "Playbook is reliable," and "Playbook provides security." These items were removed, and the mean of the remaining nine items was computed to form *trust_trimmed*.

Kendall's tau correlations between *trust_trimmed* and the six UEQ subscales are reported in Table 2. Post-mission trust was significantly positively correlated with the Dependability ($\tau = .61$, $p = .0045$, p-FDR = .0158), Efficiency ($\tau = .60$, $p = .0053$, p-FDR = .0158), and Attractiveness ($\tau = .55$, $p = .0099$, p-FDR = .0198) subscales of the UEQ. The remaining subscales were not significantly correlated with trust after FDR correction (Figure 2).

Table 2. Kendall's Tau Correlations Between UEQ Subscale Scores and *trust_trimmed* Scores ($n = 15$).

UEQ dimension	τ	*p*	*p*-FDR
Dependability	.61	.0045	**.0158**
Efficiency	.60	.0053	**.0158**
Attractiveness	.55	.0099	**.0198**
Novelty	.42	.0445	.0668
Perspicuity	.33	.1242	.1490
Stimulation	.19	.3830	.3830

[a]Bolded values indicate significant p-values after correction.

Conclusion

We used the unique opportunity afforded by a 45-day simulated spaceflight mission to address two questions: How does trust in a spaceflight-relevant scheduling software tool change over time? And how do standardized measures of trust and usability relate to one another? Our analyses showed a modest yet reliable increase in trust in the software tool over the course of the mission. An item-level analysis showed that this increase was driven by perceptions of security and familiarity. Trust was also positively correlated with multiple dimensions of usability as gauged by the UEQ, replicating Marquez et al. (2022). Even after trimming trust items that were semantically redundant with those of the UEQ's Dependability subscale, significant coupling emerged between trust and the perceived Dependability, Efficiency, and Attractiveness of Playbook.

Our work informs theoretical models of human-system trust by empirically demonstrating that trust is an evolving, multifaceted construct in an operational spaceflight setting (de Visser et al., 2020; Schaefer et al., 2016). Significant increases in security and familiarity items suggest that the general increase in trust over time was driven by shifting perceptions of security the tool provides and accumulated experience with the tool rather than by a homogenous increase across items. The general pattern observed corroborates recent work in space analog environments (Vanderdonckt et al., 2025). We also generally align with previous work showing that different dimensions of trust change over time (Chancey et al., 2025), which further demonstrates the benefit of assessing the sensitivity of the "ingredients" of singular overall trust composites.

The positive relationship between trust and usability at the end of the mission suggests that Marquez et al.'s (2022) original finding was not merely an artifact of overlapping item wording. The fact that trust/UEQ convergence replicated in this more ecologically valid setting is compelling. Drawing on this, we speculate that iterative design approaches focused on improving user experience, such as by tracking responses

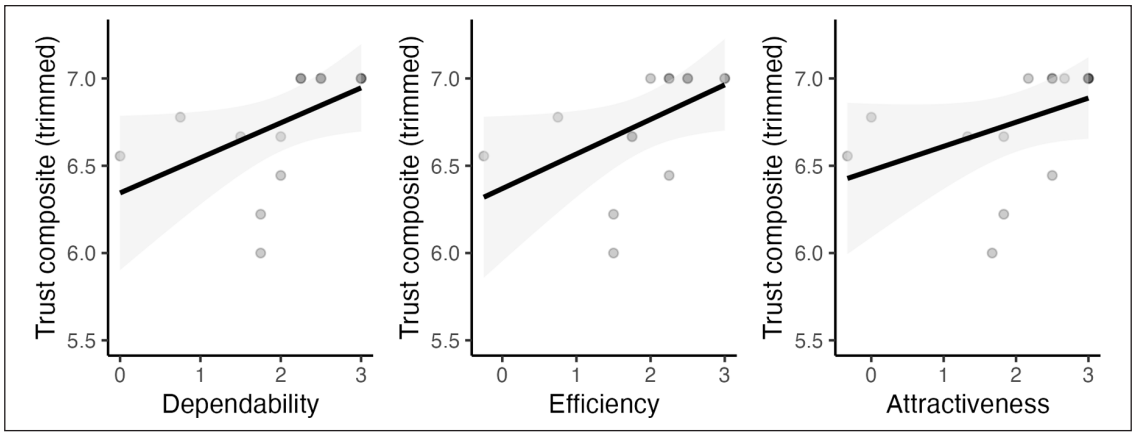


Figure 2. Positive correlations between *trust_trimmed* and the UEQ Dependability, Efficiency, and Attractiveness subscales. Points represent individual participants. Error shading represents the 95% confidence interval.

to surveys like the UEQ, may also inadvertently contribute to creating software that users find more trustworthy.

Changes in trust over time could themselves be valuable indicators of HCI quality. Trust should not decline over time if a system supports users effectively. Consider a tool with an elegant, aesthetic design that elicits early feelings of trust, only for users to discover over repeated interactions that it works poorly. In such a case, the initial illusory sense of trust would decrease over time as users realize the tool's shortcomings. Framed within our data, we can reasonably infer that Playbook successfully supported mission operations, building trust over time. We recommend that HCI practitioners measure trust repeatedly to allow such inferences.

Limitations and Future Work

Several limitations of this work should be acknowledged. First, the sample size is extremely small ($n = 16$; $n = 15$ for post-mission correlations). This is an inherent constraint of spaceflight analogs, which are time-consuming and costly to conduct, and which draw from a selective participant pool. Consequently, the statistical power and generalizability of our analyses are limited.

Second, trust was measured at only three time points over the course of the mission. While this was sufficient to establish a general trend, denser sampling would increase power and allow us to characterize trust's fluctuations with finer granularity. In a similar vein, future work would benefit from longitudinally measuring both trust *and* usability. This would better capture their evolution over time. Denser sampling would enable tests of whether early usability perceptions predict later trust, or whether trust emerges only after extended interaction.

Third, a small subset of participants showed decreases in trust over time. Targeted interviews with these crew members or deeper analyses of their tool interaction data might shed light on why this was the case. The temporal dynamics of trust at the individual level could serve as a diagnostic tool to trigger such follow-ups to better understand the data. This approach should be leveraged and validated in future research.

Fourth, our item-level analysis of Jian et al.'s (2000) measure yielded interesting insights, but it may be better to implement a validated measure that breaks trust into discrete subscales (Chancey et al., 2025; Gulati et al., 2019; Madsen & Gregor, 2000). This would be analogous to how the UEQ divides usability into Attractiveness, Perspicuity, Efficiency, Dependability, Stimulation, and Novelty. In this spirit, prior work suggests that multidimensional HCI measures can offer richer insights than unidimensional ones (Shelat et al., 2025).

Fifth, it would be beneficial to examine which objective behavioral metrics are mediated by changes in trust over time. One candidate metric is the frequency or duration of Playbook usage. Another candidate metric is how often users implement some of Playbook's automated features, such as suggested fixes. In the future, detailed analysis of interaction logs could complement the trust analyses presented here.

In summary, attending to the temporal dynamics of trust and its relationship with usability enables meaningful inferences about the quality of human-computer interaction. While no NASA standard exists for trust, high usability is required; ensuring usability may capture other desirable software qualities like promoting trust. As intelligent technologies become embedded across domains on Earth and beyond, it is essential to carefully study human perceptions of software in high-fidelity environments.

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